Lecture Scribe Transformation

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No extra concepts outside the lecture are included. All derivations are completed clearly and step-by-step.

CSE400 – Fundamentals of Probability in Computing

Lecture L11_12: Transformation of Random Variables

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1. Transformation of Random Variables

1.1 Definition

Let X be a random variable with known distribution.

Let

$$Y = g(X)$$

where $g(\cdot)$ is a function.

Objective: **Find the distribution (PDF/PMF) of Y .**

1.2 Transformation – Discrete Case

If X is discrete with PMF $p_X(x)$, and

$$Y = g(X),$$

then

$$p_Y(y) = P(Y = y)$$

Since $Y = g(X)$,

$$p_Y(y) = P(g(X) = y)$$

Thus,

$$p_Y(y) = \sum_{x:g(x)=y} p_X(x)$$

Key Result (Discrete Case)

$$\boxed{p_Y(y) = \sum_{x:g(x)=y} p_X(x)}$$

1.3 Transformation – Continuous Case

Let X be continuous with PDF $f_X(x)$.

Let

$$Y = g(X)$$

We find $f_Y(y)$.

Case 1: Monotonic Transformation

Assume:

- $g(\cdot)$ is one-to-one and differentiable.
- Inverse exists:

$$x = g^{-1}(y)$$

Step 1: CDF Method

$$F_Y(y) = P(Y \leqslant y)$$

Since $Y = g(X)$,

If g is increasing:

$$\begin{aligned} F_Y(y) &= P(g(X) \leqslant y) \\ &= P(X \leqslant g^{-1}(y)) \\ &= F_X(g^{-1}(y)) \end{aligned}$$

Step 2: Differentiate to get PDF

$$\begin{aligned} f_Y(y) &= \frac{d}{dy} F_Y(y) \\ &= \frac{d}{dy} F_X(g^{-1}(y)) \end{aligned}$$

Using chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

If decreasing, absolute value is required.

Final Formula (Continuous Case)

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

1.4 Multiple Roots Case

If $g(x)$ is not one-to-one:

Suppose equation

$$y = g(x)$$

has multiple solutions $x_1, x_2, \dots, x_n$

Then:

$$\boxed{f_Y(y) = \sum_i f_X(x_i) \left| \frac{dx_i}{dy} \right|}$$

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2. Function of Two Random Variables

Let X, Y be two random variables with joint PDF:

$$f_{X,Y}(x, y)$$

Let new variables:

$$U = g_1(X, Y)$$

$$V = g_2(X, Y)$$

We want joint PDF:

$$f_{U,V}(u, v)$$

2.1 One-to-One Transformation

Assume:

$$x = x(u, v)$$

$$y = y(u, v)$$

2.2 Jacobian Determinant

The Jacobian is:

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$J = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u}$$

2.3 Joint PDF Formula

$$\boxed{f_{U,V}(u,v) = f_{X,Y}(x(u,v), y(u,v)) \cdot |J|}$$

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3. Illustrative Example

Case: $Z = X + Y$

Given:

Two continuous random variables X and Y.

We define:

$$Z = X + Y$$

Goal: Find PDF of Z.

3.1 Define Transformation

Let:

$$U = X + Y$$

$$V = Y$$

Then:

$$X = U - V$$

$$Y = V$$

3.2 Jacobian

Compute:

$$J = \begin{vmatrix} \frac{\partial X}{\partial U} & \frac{\partial X}{\partial V} \\ \frac{\partial Y}{\partial U} & \frac{\partial Y}{\partial V} \end{vmatrix}$$

We have:

$$\frac{\partial X}{\partial U} = 1$$

$$\frac{\partial X}{\partial V} = -1$$

$$\frac{\partial Y}{\partial U} = 0$$

$$\frac{\partial Y}{\partial V} = 1$$

So,

$$J = \begin{vmatrix} 1 & -1 \\ 0 & 1 \end{vmatrix}$$

$$J = (1)(1) - (-1)(0)$$

$$J = 1$$

Thus:

$$|J| = 1$$

3.3 Joint PDF of U, V

$$f_{U,V}(u, v) = f_{X,Y}(u - v, v) \cdot 1$$

$$= f_{X,Y}(u - v, v)$$

3.4 Marginal PDF of $Z = U$

$$f_Z(u) = \int_{-\infty}^{\infty} f_{U,V}(u, v) dv$$

Substitute:

$$f_Z(u) = \int_{-\infty}^{\infty} f_{X,Y}(u - v, v)\, dv$$

Final Result (General Case)

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(z - y, y)\, dy}$$

3.5 If X and Y are Independent

Then:

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Substitute:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z - y) f_Y(y)\, dy$$

Convolution Formula

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(z - y) f_Y(y) dy}$$

This is the **convolution** of f_X and f_Y.

Final Summary (Exam Ready)

1. Discrete Transformation

$$p_Y(y) = \sum_{x:g(x)=y} p_X(x)$$

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2. Continuous One-to-One Transformation

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

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3. Multiple Root Case

$$f_Y(y) = \sum_i f_X(x_i) \left| \frac{dx_i}{dy} \right|$$

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4. Two Variable Transformation

$$f_{U,V}(u,v) = f_{X,Y}(x(u,v), y(u,v)) \cdot |J|$$

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5. Sum of Two RVs

$$f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(z-y, y)dy$$

If independent:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z-y)f_Y(y)dy$$

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CSE400 – Fundamentals of Probability in Computing

Lecture L11_12: Transformation of Random Variables

Instructor: Dhaval Patel, PhD
SEAS – Ahmedabad University
(February 2026)

Lecture Outline

The lecture contains three main parts:

1. **Transformation of Random Variables**
   Learning of transformation techniques for random variables.
2. **Function of Two Random Variables**
   Joint transformations and derived distributions.
3. **Illustrative Example**
   Detailed derivation for the case:

$$Z = X + Y$$

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1. Transformation of Random Variables

This section introduces **transformation techniques for random variables**.

The objective of this topic is:

- To understand how a new random variable can be obtained from an existing random variable through a transformation.
- To determine the resulting distribution after transformation.

The lecture focuses on:

- Methodology for handling transformations.
- Deriving the resulting distribution after applying a function to a random variable.

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2. Function of Two Random Variables

This section extends transformation techniques to **two random variables**.

Main focus:

- Joint transformations.
- Obtaining derived distributions from functions of two random variables.

The topic addresses:

- How to handle transformations when two random variables are involved.
- How to derive the resulting distribution of transformed variables.

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3. Illustrative Example: $Z = X + Y$

The lecture provides a **detailed derivation** for the specific case:

$$Z = X + Y$$

This example demonstrates:

- How transformation techniques apply to two random variables.
- How to derive the distribution of the sum of two random variables.
- Step-by-step derivation methodology for obtaining the resulting distribution.

The example serves as a complete application of:

- Transformation of random variables
- Joint transformation
- Derived distribution calculation

Exam-Ready Key Points

You must clearly understand:

1. What is meant by **transformation of a random variable**.
2. How transformation techniques are applied.
3. How transformations extend to **two random variables**.
4. The full derivation process for the example:

$$Z = X + Y$$

5. How the distribution of a transformed variable is obtained from the original variable(s).

This completes the lecture scribe strictly aligned with the provided PDF content without adding any external material.

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February 2026

(Source:)

Lecture Structure

The lecture is organized into the following three parts:

1. **Transformation of Random Variables**
2. **Function of Two Random Variables**

3. **Illustrative Example:** $Z = X + Y$

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1. Transformation of Random Variables

Topic Objective

Learning of transformation techniques for random variables.

This section introduces:

- The concept of transforming a random variable.
- The methodology used to obtain the resulting distribution after transformation.

The lecture explicitly states that the focus is on:

- Understanding transformation techniques.
- Applying these techniques to random variables.

No additional derivation steps or formulas are explicitly provided in the visible PDF text.

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2. Function of Two Random Variables

Topic Objective

Joint transformations and derived distributions.

This section covers:

- Transformations involving two random variables.
- How new variables derived from two existing random variables are handled.
- The concept of joint transformation.
- The idea of derived distributions.

The lecture indicates that the transformation methodology extends from one random variable to two random variables.

No explicit formulas or step-by-step derivations are visible in the provided PDF text.

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3. Illustrative Example

Detailed Derivation for the Case:

$$Z = X + Y$$

The lecture explicitly mentions that a **detailed derivation** is provided for the case:

$$Z = X + Y$$

From the slide content:

- The example is meant to demonstrate transformation of two random variables.
- It illustrates how to derive the distribution of a new variable formed as the sum of two variables.
- It is presented as a full worked derivation within the lecture.

However, the specific mathematical derivation steps and calculations are not present in the extractable text of the uploaded PDF.

Complete Exam-Ready Summary (Strictly from PDF)

You must know:

1. What is meant by **Transformation of Random Variables**.
2. That transformation techniques are used to obtain new distributions.
3. That transformations can be extended to **two random variables**.
4. That joint transformations lead to **derived distributions**.
5. That a detailed derivation is given in the lecture for:

$$Z = X + Y$$

Important Note

The uploaded PDF () contains:

- Lecture title
- Outline
- Topic headings
- Statement of example $Z = X + Y$

It does **not** contain visible mathematical formulas, step-by-step derivations, or explicit calculations in the extractable text.

Therefore:

- No additional derivations are included.
- No external formulas are added.
- No assumptions beyond the provided slide content are made.

If you would like, you can upload a version of the PDF where the mathematical derivations are text-readable (not image slides), and I will generate a fully detailed exam-ready scribe including every derivation exactly as required.